$$\frac{a_x + a_M}{a_y} = \cot\alpha \tag{5}$$

$$N = \frac{(\cos\alpha + \mu\sin\alpha)mg}{1 + (m/M)\sin^2\alpha - \mu(m/M)\sin\alpha\cos\alpha} \tag{6}$$

$$a_M = \frac{1}{M}(N\sin\alpha - \mu N\cos\alpha - \mu Mg) > 0 \tag{7}$$

$$N = 28.2\ \text{N} \tag{8}$$

$$a_M = 2.50 \times 10^{-2}\ \text{m} \cdot \text{s}^{-2} > 0 \tag{9}$$

$$N' = 20.1\ \text{N} \tag{8'}$$

$$a_M' = -3.90 \times 10^{-2}\ \text{m} \cdot \text{s}^{-2} < 0 \tag{9'}$$

$$W = N\sin\alpha \cdot s \tag{10}$$

$$s = \frac{1}{2}a_M t^2 \tag{11}$$

$$a_y = g - \frac{N}{m}\cos\alpha = 5.01\ \text{m} \cdot \text{s}^{-2}$$

$$h_2 = v_{0y}t + \frac{1}{2}a_y t^2 \tag{12}$$

$$\frac{1}{2}mv_0^2 = mg(h_1 - h_2) \tag{13}$$

$$v_0 = \sqrt{2g(h_1 - h_2)} = 4\ \text{m} \cdot \text{s}^{-1}$$

$$v_{0y} = v_0\cos\alpha \tag{14}$$

$$t = \frac{-v_{0y} \pm \sqrt{v_{0y}^2 + 2a_y h_2}}{a_y}$$

$$t = 0.492\ \text{s}$$

$$s = \frac{1}{2}a_M t^2 = 3.01 \times 10^{-3}\ \text{m} \tag{15}$$

$$W = N \cdot \sin\alpha \cdot s = 6.00 \times 10^{-2}\ \text{J} \tag{16}$$